

Jan Sobotka

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EDUCATION

Swiss Federal Institute of Technology in Lausanne (EPFL)

Sep. 2024 – ongoing

M.S. in Computer Science

Lausanne, CH

- Weighted grade average: **5.89** (scale of 1.0 to 6.0, 6.0 is the highest)
- Anticipated graduation date: Aug. 2026

Czech Technical University, Faculty of Information Technology

July 2021 – June 2024

B.S. in Informatics, Specialization in Artificial Intelligence

Prague, CZ

- Weighted grade average: **1.05** (scale of 1.0 to 4.0, 1.0 is the highest)

RESEARCH EXPERIENCE

Research Assistant at the Autonomous Systems Group

July 2025 – ongoing

University of Texas at Austin, Oden Institute for Computational Engineering and Sciences

Austin, US

- Analyzing large language models in strategy games using mechanistic interpretability and game theory.
- Supervised by Prof. Ufuk Topcu.

Research Assistant at the MLBio Lab

Aug. 2024 – July 2025

Swiss Federal Institute of Technology in Lausanne (EPFL)

Lausanne, CH

- Enhanced *Weak-to-Strong generalization* (learning from weaker teachers) by dynamically reweighting teaching signals, improving robustness under distribution shifts by over 30% [2].
- Evaluated in-context learning capabilities of multimodal foundation models on computer vision tasks.
- Supervised by Prof. Maria Brbić.

Research Intern at the Computational Systems Neuroscience Group

Sep. 2023 – Aug. 2024

Faculty of Mathematics and Physics, Charles University

Prague, CZ

- Developed a state-of-the-art method for decoding images from neural population activity [1].
- Supervised by Prof. Ján Antolík.

Computational Neuroscience Research Intern

July 2023 – Sep. 2023

Biozentrum, University of Basel

Basel, CH

- Designed computational models of spiking neural networks and applied dynamical systems theory to demonstrate how bistable neuronal dendrites enable a 1,000-fold increase in memory capacity.
- Supervised by Dr. Everton J. Agnes.

Research Assistant at the Data Science Lab

Apr. 2023 – Feb. 2024

Faculty of Information Technology, Czech Technical University

Prague, CZ

- Analyzed training dynamics of *Learning-to-Optimize* (meta-learning) [6,7] and *fractional gradient descent* [5].

PUBLICATIONS

* DENOTES EQUAL CONTRIBUTION

- [1] Do LLMs Strategically Reveal, Conceal, and Infer Information? A Theoretical and Empirical Analysis in The Chameleon Game. [PDF] 
Mustafa O. Karabag, **Jan Sobotka**, Ufuk Topcu.
Conference on Autonomous Agents and Multiagent Systems (AAMAS 2026, Extended Abstract).
- [2] MEIcoder: Decoding Visual Stimuli from Neural Activity by Leveraging Most Exciting Inputs. [PDF] 
Jan Sobotka, Luca Baroni, Ján Antolík.
Neural Information Processing Systems (NeurIPS 2025).
- [3] Weak-to-Strong Generalization under Distribution Shifts. [PDF] 
Myeongho Jeon*, **Jan Sobotka***, Suhwan Choi*, Maria Brbić.
Neural Information Processing Systems (NeurIPS 2025).

- [4] Reverse-Engineering Memory in DreamerV3: From Sparse Representations to Functional Circuits. [PDF 
Jan Sobotka, Auke Ijspeert, Guillaume Bellegarda.
Neural Information Processing Systems (NeurIPS 2025, Spotlight at Mech Interp Workshop).
- [5] Enhancing Fractional Gradient Descent with Learned Optimizers. [PDF 
Jan Sobotka, Petr Šimánek, Pavel Kordík.
ArXiv preprint (2025).
- [6] Investigation into the Training Dynamics of Learned Optimizers. [PDF 
Jan Sobotka, Petr Šimánek, Daniel Vašata.
International Conference on Agents and Artificial Intelligence (ICAART 2024).
- [7] Investigation into the Training Dynamics of Learned Optimizers (Student Abstract). [PDF 
Jan Sobotka, Petr Šimánek.
AAAI Conference on Artificial Intelligence (AAAI 2024).

INDUSTRY EXPERIENCE

- Research Data Scientist** Feb. 2026 – ongoing
AXA Group Operations, AI Research & Development Paris, FR
• Conducting a Master's thesis project titled “Understanding the Development of Reasoning in Language Models”.
- AI/ML Engineer Junior** June 2021 – June 2023
Generali Česká pojišťovna Prague, CZ
• Prepared a machine learning pipeline that improved the accuracy of product recommendations by 35%.
• Built computing infrastructure for the company's internal data science community with over 90 members.
• Applied deep learning to product and text recommendation, client departure prediction, email classification, and unsupervised customer segmentation.

SELECTED PROJECTS

- Deep Reinforcement Learning for Optimal Experimental Design in Biology** Jan. 2023 – June 2023
• Open-science project focused on efficient estimation of biological system parameters [OpenBioML 

Generative Models of Regulatory DNA Sequences Based on Diffusion Models July 2022 – Dec. 2022
• Open-science project investigating the application of diffusion models to genomics data [OpenBioML 

EXTRACURRICULAR ACTIVITIES

- Organizer of the Traion Community of Student Entrepreneurs** June 2020 – Feb. 2021
• Organized offline meetings, educational seminars, and workshops for student entrepreneurs.
- Volunteer for an Entrepreneurship Education Program for Students** Dec. 2019 – Aug. 2020
• Organized events and wrote a technology/entrepreneurship blog for the Soutěž and Podnikaj organization.
- Pitcher at the Czech Republic National Baseball Team U-15** Jan. 2017 – July 2017
• Secured third place at the U-15 European Baseball Championship 2017.

HONORS AND AWARDS

The Bakala Foundation Scholarship: Awarded to top 7% of applicants (academics & extracurriculars) | 2024
Merit-Based Scholarship, Czech Technical University: Awarded for a high academic average | 2021, 2022, 2023
National Benchmark Exam in Mathematics SCIO : Scored higher than 97% of the 875 test takers | 2021
Algorithms & Programming Competition FIKS: Ranked 4th out of 107 participants | 2020
TOP25 Czech High School Students of the Year 2020: Selected based on extracurricular activities | 2020

SKILLS

Programming languages: Python, C, C++, Julia, JavaScript, Go
Other selected technologies: PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, Azure, AWS, Git, Docker, Bash
Languages: Czech (native), English (C1, TOEFL iBT 105), German (A2)